DIGITAL TWIN HACKATHON

The Smart Companion

An AI assistant that turns big goals into small wins, for neurodivergent users, now extended into healthcare

TEAM LEADER
Anwesha Mondal

MEMBER 02
Sandipan Dafadar

MEMBER 03
Meghali Khan

COLLEGE Techno India University, Kolkata, WB

THE PROBLEM

Why big tasks feel impossible

01

Starting is hard

People with ADHD and other neurodivergent conditions struggle with task initiation and decision paralysis

02

Text overwhelms

Normal to-do apps and chatbots give long lists and walls of text, which overwhelms them

03

Health is harder

Health management (medicines, readings, emergencies) is also hard for them, and for elderly and rural users

THE NEED

One simple, friendly, privacy-safe assistant that guides, sees, listens and helps in emergencies

Project objectives

01 Break any goal into 3 small steps at a time (micro-wins)

02 Remember the conversation and keep topics in separate chats

03 Support voice, photos, live camera / video streaming and PDFs

04 Protect privacy by redacting personal data before AI processing

05 Add motivation: streaks, badges, progress

06 Healthcare support: doctor directory, emergency triage, SOS, health log

07 Use only free, open tools so it is affordable

Small steps. Big wins.

UNDER THE HOOD

Architecture and tech stack

Frontend

No build step

- HTML5 · CSS3 · ES6+ JS
- Web Speech API
- MediaRecorder API
- getUserMedia camera

FastAPI backend

Python 3.10+

- Task decomposition engine
- PII redaction
- Speech-to-text client
- Vision Q&A and guided search
- IP camera stream worker
- Doctors · SOS · Health Log

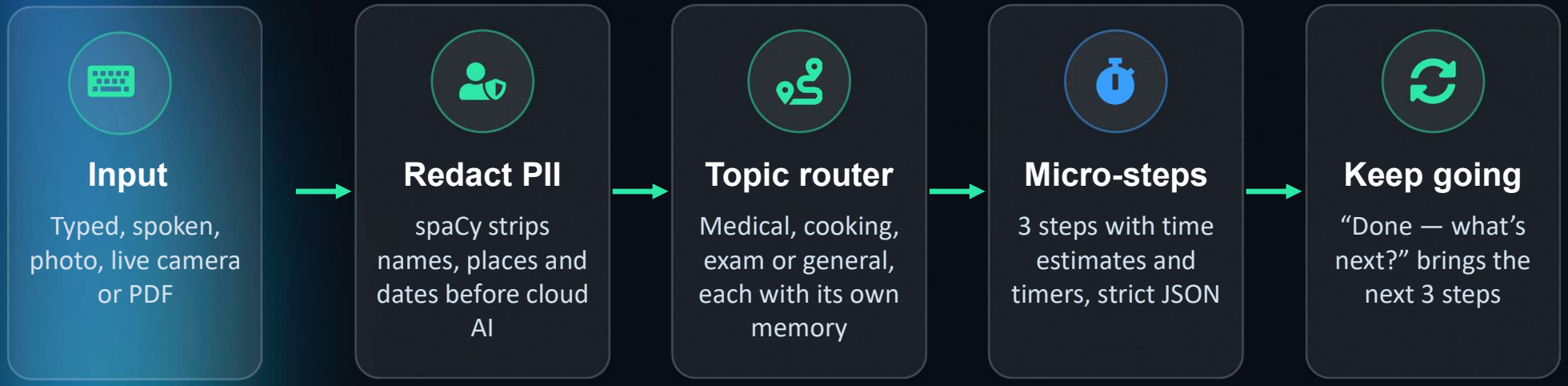
AI engines

Free and open

- Groq LLM
- Whisper large-v3-turbo
- Groq Vision
- YOLOv8n (local)
- spaCy NER (local)

HOW IT WORKS

From a big goal to the next small step

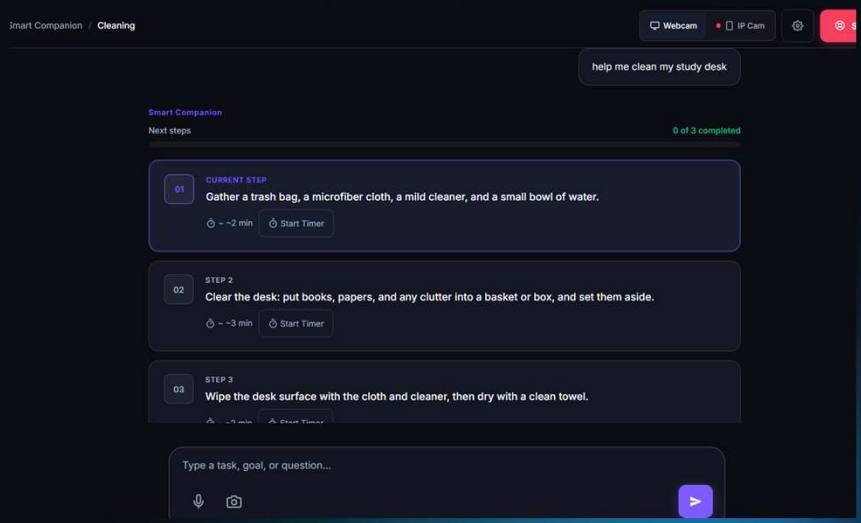


EMERGENCY LANE

Emergency detected: instant 108 and 104 guidance, SOS panel with live GPS and nearby hospitals, alert to saved contacts

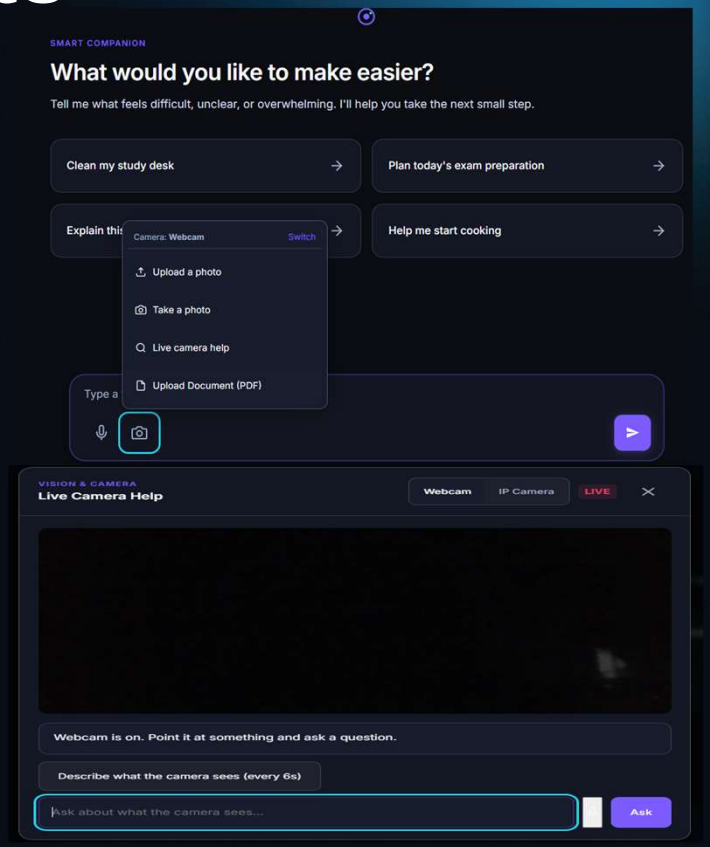
Core intelligence

- Progressive step breakdown with time estimates and timers
- Strict JSON output, retry and step trimming
- Repeat detection so tasks end properly
- Clarifying questions to avoid decision paralysis
- Topic router with separate memory per chat (medical, cooking, exam, general)
- “Explore more”: links and images for every answer



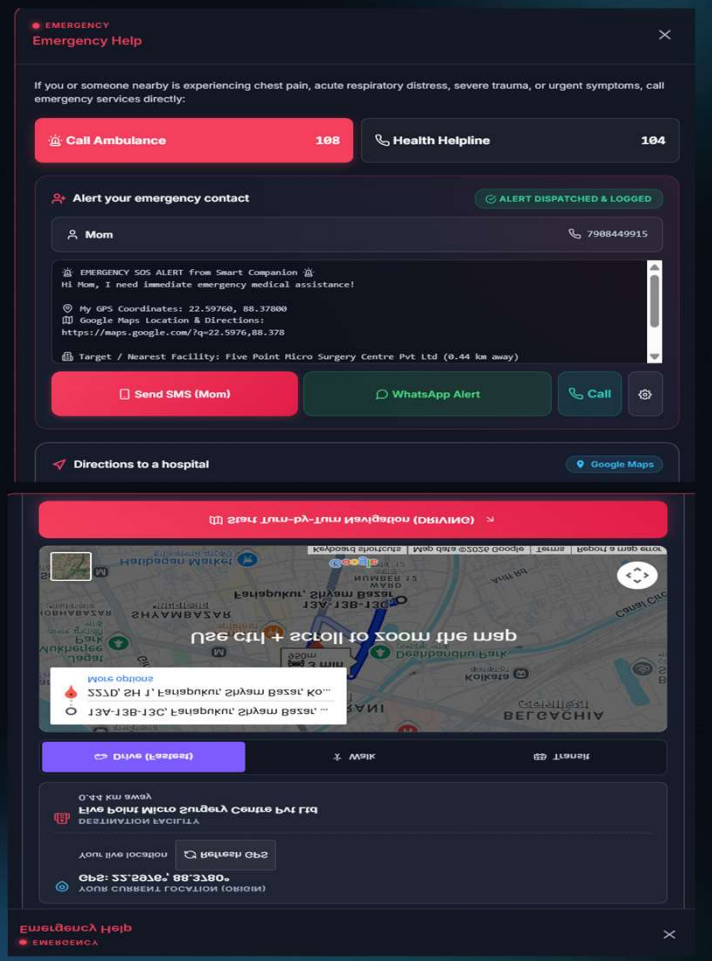
Vision, voice and documents

- Live Camera Help with webcam or phone IP camera
- Upload or take a photo and ask questions
- PDF question-answering, including scanned pages
- Voice input and read-aloud
- Ambient mode that describes the scene every 6 seconds



Healthcare and safety

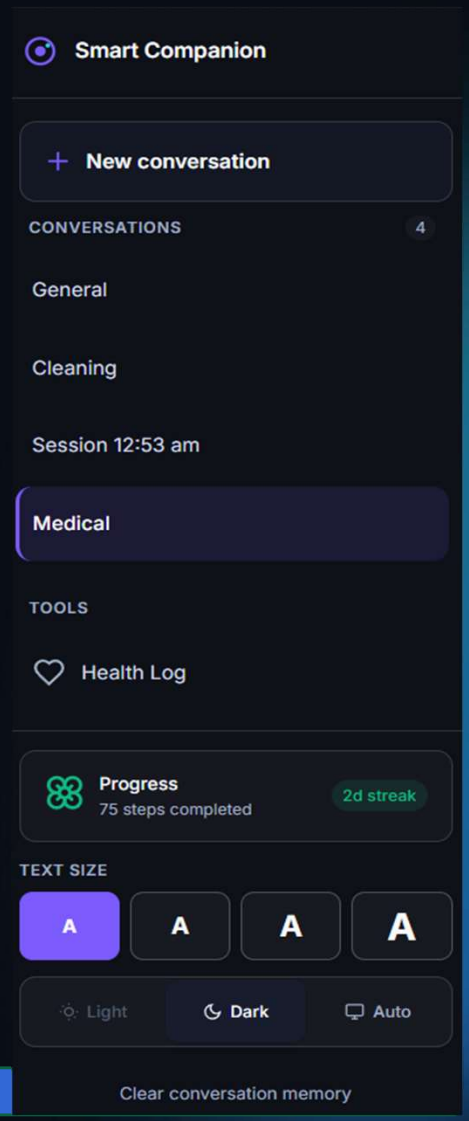
- Self-built free doctor directory (SQLite, OpenStreetMap data, verified-first ranking)
- Appointment requests sent via WhatsApp link
- Emergency detection: instant 108 and 104 guidance
- SOS panel: live GPS, nearby hospitals, map routes (drive, walk, transit)
- Ready-made alert message for saved contacts, sent by SMS, WhatsApp or call in one tap, with alerts logged
- Health Log: blood pressure, glucose and medicines



WORK COMPLETED · 4 OF 4

Neuro-inclusive design

- Side panel with four text sizes
- Light, dark and auto themes
- Gamification: progress tile, streak, badges
- One conversation per topic, with delete option
- Calm, friendly tone and one task at a time



Current progress

FEATURE	STATUS
Task breakdown, chat memory, privacy layer	Done
Voice, vision, live camera, PDFs	Done
Doctor directory, triage, SOS, Health Log	Done, under testing
Explore more (links and images)	Done

26 API endpoints

3 AI models

4 chat modes

WHAT COMES NEXT

Future plan



01

Accessibility

- Voice-only mode for blind users
- Visual, text-only mode for deaf users



02

Care and safety

- Daily wellness reminders (medicines, readings)
- Caregiver features and medicine-photo reading
- Crisis support flow with a contact the user chooses



03

Reach and trust

- Mobile app version and wider doctor coverage
- Testing with real users

IN SHORT

From big goals to small wins



Guides

Any goal becomes 3 small steps at a time, with timers, memory per chat and streaks to keep going.



Sees and listens

Voice, photos, live camera and PDFs let users speak, show and ask instead of reading walls of text.



Protects

Personal data is redacted before AI processing, and triage, SOS and a health log help when it matters.

THE SMART COMPANION

Thank you

Questions?

